Talking Machines: A Fixture Paper

1. Introduction

Speech synthesis is the artificial production of human speech. A computer system used for this purpose is called a speech synthesizer, and it can be implemented in software or in hardware. A text-to-speech system converts normal language text into speech; other systems render symbolic linguistic representations like phonetic transcriptions into speech.

Synthesized speech can be created by concatenating pieces of recorded speech that are stored in a database. Systems differ in the size of the stored speech units; a system that stores phones or diphones provides the largest output range, but may lack clarity. For specific usage domains, the storage of entire words or sentences allows for high-quality output. Alternatively, a

synthesizer can incorporate a model of the vocal tract and other human voice characteristics to create a completely synthetic voice output.

2. History

Long before the invention of electronic signal processing, some people tried to build machines to emulate human speech. Some early legends of the existence of speaking heads involved Gerbert of Aurillac, Albertus Magnus and Roger Bacon.

• The acoustic-mechanical speech machine of Wolfgang von Kempelen.
• The Voder, exhibited at the 1939 New York World's Fair.

3. Conclusion

Modern systems use neural networks to generate the waveform directly from text, and they read news, books and screens aloud.

References

[1] A reference entry that must not be read aloud.